

SHRIVATHSA M S

Full Stack Engineer

Full-stack engineer with a systems and robotics background

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KEY TECHNICAL SKILLS

Programming Languages:	Python, TypeScript, JavaScript, Go, C++
Frontend:	React.js, Next.js, Redux, Webpack, Web Audio API
CSS & UI Libraries:	Tailwind CSS, Chakra UI, Material Design, SASS
Backend:	FastAPI, Node.js, Express.js, REST API Design, WebSockets, Microservices
Databases:	PostgreSQL, MongoDB, SQLite
Infra & Tools:	Docker, Linux, AWS (EC2, S3), Cloudflare R2, Railway, GitHub Actions
Testing:	Jest, Cypress, Vitest, pytest
SDLC:	Agile, Git, Jira, Gerrit

EXPERIENCE

Solvendo - Legal-tech startup

Software Engineer

Bengaluru, KA

July 2023 - February 2025

- Built and shipped frontend features for a bankruptcy resolution platform used by institutions including Canara Bank and IBBI, handling complex multi-role workflows across banks, companies, and legal teams
- Bootstrapped a new Next.js frontend from scratch, establishing coding standards and a reusable component library that the team built features on top of. Technologies used: Next.js, Chakra UI, NextAuth.js
- Built a high-performance blockchain analytics dashboard with rich interactive time-series charts and a drag-and-drop analysis interface — sub-2s load time and a Lighthouse score of 93 at 10k MAU. Technologies used: Next.js, React-DND, ApexCharts
- Diagnosed and resolved React performance bottlenecks using React DevTools and Lighthouse, cutting First Contentful Paint by 50%
- Built a blockchain data aggregation microservice in Go using WebSockets and REST endpoints, achieving 8x higher throughput than the previous solution. Technologies used: Go, net/http, WebSockets

Indian Institute of Science

Research Engineer

Bengaluru, KA

January 2023 - July 2023

- Engineered multi-agent robot systems for agricultural applications, achieving 2x improvement in obstacle avoidance performance through novel algorithms. Technologies used: Python, ROS, EGO Planner
- Developed motion planning and perception software for UAVs in Python. Technologies used: ROS, Intel Realsense cameras, 2D Lidars
- Integrated UAVs with DJI control software using SDKs to enhance automation capabilities. Technologies used: Python, DJI SDK

NewSpace Research & Technologies - UAV & robotics R&D company

Robotics Engineer

Bengaluru, KA

October 2021 - January 2023

- Worked on ROS-based autonomy software for UAVs operating in GPS-denied environments. Technologies used: ROS, Python, Linux, C++
- Optimized robot localization using visual SLAM and LiDAR SLAM algorithms in C++. Technologies used: OpenCV, OpenSLAM, VINS-Stereo/Mono, Ouster Lidars
- Developed UAV simulation tools using Python and physics simulation software. Technologies used: Python, Gazebo, Docker, C++

PROJECTS

CoeMix *Python, FastAPI, React, TypeScript, PostgreSQL, Cloudflare R2, Railway*

A mix comparison tool for audio engineers. Validates and processes uploaded audio server-side using fprobe and threaded analysis, gain-matches tracks via integrated LUFs, and computes loudness, frequency, and spatial features. Frontend renders results through meters and graphs, as well as a live spectrum and goniometer via the Web Audio API. [Live Demo](#)

BarFly Browser Extension *TypeScript, Firefox/Chrome Browser Extensions API*

A browser extension that replaces the bookmarks toolbar with an LRU-cached view of recently interacted bookmarks, keeping the most relevant links always accessible. [Firefox store page](#)

WikiTree Browser Extension *TypeScript, React.js, Chrome Browser Extensions API*

A Chrome extension to visualize Wikipedia rabbit holes as interactive trees, built using TypeScript and React. [Chrome store page](#)

EDUCATION

BMS College of Engineering

Degree in Mechanical Engineering (BE)

Electives: AI, Java, Robotics *GPA: 8.75*

Bengaluru, KA

July 2017 - July 2021

PUBLICATIONS

Kumar Ankit, **Shrivathsa M S**, Shishir Kolathaya, Debasish Ghose. "Reachability Analysis of UAVs through Cluttered Environments." *AIAA SciTech 2024 Forum*, January 2024. [DOI: 10.2514/6.2024-2017](#)